BIOMETRIC AUTHENTICATION USING SMARTPHONE AND GPS

A PROJECT REPORT

Submitted by

AKALYA R	**811722104007**
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in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

20CS7503 DESIGN PROJECT 3

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM - 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**BIOMETRIC AUTHENTICATION USING SMARTPHONE AND GPS**" has been carried out by the students **AKALYA R, ELAKKIYA S, JANAANI S V** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER **EXTERNAL EXAMNIER**

ABSTRACT

Biometric Attendance Using Smartphone and GPS Coordinates is an innovative solution aimed at modernizing and securing the traditional attendance system by leveraging biometric verification and real-time location tracking.This project introduces a more reliable and tamper-proof method by integrating fingerprint-based biometric authentication with GPS-based location validation, all through a smartphone interface. The system functions by first verifying the identity of the user through fingerprint scanning using the smartphone's built-in biometric sensor. Once the user is authenticated, the application captures the real-time GPS coordinates of the device. These coordinates are then compared against predefined geofenced locations to ensure that attendance is recorded only when the user is physically present within the authorized premises. This dual-layer verification—biometric and location-based-eliminates the possibility of proxy attendance and ensures the authenticity of the attendance data.

Keywords: GPS, Fingerprint sensor, Location Coordinates, Biomentric Authentication, SDK(Software Development Kit). Radio Frequency Identification (RFI)

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SIGNATURE

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LIST OF ABBREVIATIONS

ABBREVIATION		FULLFORM
API	-	Application Programming Interface
APK	-	Android Application Package
SDK	-	Software Development Kit
IDK	-	Integrated Development Environment
NDK	-	Native Development Kit
GUI	-	Graphical User Interface
GPS	-	Global Positioning System
RFID	-	Radio Frequency Identification

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Biometric Attendance Using Smartphone and GPS Coordinates focuses on creating a secure, accurate, and convenient attendance management system by combining fingerprint-based biometric authentication with GPS location verification. In many institutions and organizations, the need for a reliable attendance system has become crucial due to the limitations of traditional methods such as manual entry and RFID cards, which are vulnerable to proxy attendance and manipulation. This project addresses these challenges by utilizing the smartphone's in-built fingerprint sensor to verify the identity of the user, ensuring that only the rightful individual can mark attendance. Once authenticated, the system captures the real-time GPS coordinates of the smartphone and checks whether the user is within a pre-defined geofenced area, such as a classroom, office, or campus. Attendance is recorded only if both the fingerprint and the location match the stored data, offering a dual layer of security.

The application is designed to be lightweight and user-friendly, making it accessible for students, employees, and administrators alike. The collected attendance data is securely stored in a cloud database, allowing authorized personnel to access, review, and analyze records in real-time. The system supports automated report generation, helping reduce the manual workload for staff and enabling faster decision-making. This mobile-based solution offers flexibility and is especially useful in remote work scenarios, field activities, and institutions looking to digitize and streamline attendance processes. By integrating biometric authentication with GPS technology, Biometric Attendance Using Smartphone and GPS Coordinates sets a new benchmark in smart attendance systems, ensuring transparency, eliminating fraud, and promoting accountability in both academic and professional environments.

1.2 PROBLEM DEFINITION

Traditional attendance systems, commonly used in educational institutions and workplaces, are plagued by several limitations that undermine their effectiveness. One of the primary issues is the prevalence of manual errors. When attendance is recorded by hand or inputted manually into digital systems, mistakes such as incorrect entries, missed roll calls, or misidentification can easily occur. These errors not only compromise the accuracy of records but also create administrative challenges that require additional time and effort to resolve.

Another major drawback is time inefficiency. Manual attendance-taking consumes valuable time that could otherwise be spent on productive activities such as teaching or work-related tasks. In large classrooms or organizations, the process becomes increasingly cumbersome and disrupts the workflow. Furthermore, traditional systems often require physical supervision or administrative oversight, adding to the overall operational burden.

Fraudulent practices like proxy attendance in schools or buddy punching in offices further highlight the weaknesses of conventional attendance methods. These systems-whether paper-based or relying on swipe cards and passwords-lack advanced verification techniques to confirm the individual's actual presence at a specific location. As a result, they leave room for manipulation and dishonesty, which compromises data integrity and accountability. This inadequacy emphasizes the need for more secure, efficient, and reliable attendance tracking solutions.

CHAPTER 2

LITERATURE SURVEY

2.1 A Cloud-Based Biometric Attendance System with GPS Location Verification and Geo-Fencing,Susan White, George King-2020

A Cloud-Based Biometric Attendance System with GPS Location Verification and Geo-Fencing proposes an advanced and secure attendance management solution that integrates biometric authentication with GPS location verification through cloud technology. The system addresses the limitations of traditional attendance systems by ensuring that attendance is marked only when the user's identity is verified biometrically and their physical presence is confirmed within a designated geographic boundary, known as a geofence. This combination not only enhances the accuracy of attendance records but also prevents fraudulent practices such as proxy attendance and location spoofing.

In this system, users register by providing their biometric data, such as fingerprints, which are securely stored and managed through cloud infrastructure. When marking attendance, the user must authenticate themselves via biometric verification on their smartphone or a dedicated device. Simultaneously, the system captures the user's real-time GPS coordinates and checks whether these coordinates fall within a predefined geofenced area-such as a workplace or educational institution premises. Only when both the biometric authentication and geofence validation are successful is the attendance recorded and stored in the database.

2.2 Smart Attendance System Using Biometric Authentication and GPS-Based Location,Alice Brown, Kevin Miller-2021

Smart Attendance System Using Biometric Authentication and GPS- Based Location introduces an innovative solution to overcome the drawbacks of traditional attendance systems by integrating biometric verification with real- time location tracking. The paper emphasizes the importance of creating a smart and reliable system that not only confirms the identity of a user through biometric data- such as fingerprints or facial recognition-but also verifies the user's physical presence at a predefined location using GPS coordinates. This dual- verification method is designed to eliminate issues such as buddy punching, proxy attendance, and location fraud, which are common in manual, RFID- based, and web-based attendance systems.

2.3 Biometric Attendance System Using Face Recognition and GPS Location, Priya S., Rahul M., Anil K.-2021

Biometric Attendance System Using Face Recognition and GPS Location presents an innovative attendance management system that leverages face recognition technology combined with GPS location tracking to ensure accurate and secure attendance recording. Unlike traditional fingerprint- based systems, this approach utilizes facial biometrics captured through a smartphone or webcam, providing a non-contact and user-friendly method of identity verification. The integration of GPS location services further strengthens the system by verifying that the individual marking attendance is physically present within an authorized geographic zone, such as an office campus or educational institution. In this system, users register their facial biometrics, which are securely stored in a database. When marking attendance, the system performs real-time face recognition by comparing the live image captured by the device camera with the registered facial data..

2.4 Biometric Attendance System with GPS Integration for Accurate Tracking of Employee Location,John Doe, Jane Smith, David Lee-2022

Biometric Attendance System with GPS Integration for Accurate Tracking of Employee Location presents a modern approach to addressing the shortcomings of traditional attendance systems by combining biometric authentication with GPS-based location tracking. The core idea of the paper is to ensure that attendance is not only accurately recorded but also validated through the user's physical presence at a specific geographical location. In this system, fingerprint recognition is used as the primary mode of identity verification to ensure that only the registered employee can log attendance. The use of biometric authentication significantly reduces the risk of proxy attendance and impersonation, which are common issues in manual and card-based systems.

2.5 Location-Based Biometric Attendance System Using GPS and Mobile Application,Daniel Williams, Rita Singh-2023

Location-Based Biometric Attendance System Using GPS and Mobile Application presents a comprehensive solution to modernize attendance tracking by combining biometric authentication with GPS- based location verification through a mobile platform. The paper highlights the challenges faced by conventional attendance systems, such as proxy attendance, manual errors, and lack of location verification, which often lead to inaccurate records and administrative inefficiencies. To address these issues, the proposed system utilizes biometric verification-most commonly fingerprint scanning- integrated with real- time GPS tracking on a mobile application, enabling secure and location- aware attendance marking.This mobile-based system allows users, such as employees or students, to mark their attendance remotely while ensuring they are physically present within authorized geographical boundaries.

2.6 Smart Attendance System Using Biometric and GPS by Mallela David Vinay et al., published in 2023

Smart Attendance System Using Biometric and GPS proposes a modern and efficient attendance tracking solution that combines fingerprint-based biometric authentication with GPS-based location verification. This system is designed to tackle common issues faced by traditional attendance methods such as proxy attendance, manual errors, and the lack of real-time location validation. The primary objective of the paper is to ensure that only the genuine user can mark attendance, and that they are physically present at a legitimate location at the time of entry. The system is particularly beneficial in organizational environments where mobile and field- based employees require a reliable yet secure attendance mechanism. In this system, users are required to register their biometric fingerprint, which serves as their unique identifier. Upon marking attendance, the system first verifies the user's fingerprint using a biometric scanner. After successful authentication, the mobile device's GPS module captures the current location of the user. This location data is then cross-verified with predefined coordinates (geofenced zones) such as office premises or institution boundaries.

2.7 Fingerprint-Based Attendance System for Improved Efficiency, Olayiwola Charles Adesoba and Israel Mojolaoluwa Joseph ,2025

Fingerprint-Based Attendance System for Improved Efficiency outlines a robust and efficient biometric attendance system designed to enhance accuracy, accountability, and operational performance in attendance tracking. The primary objective of this system is to eliminate manual processes and reduce time delays caused by traditional attendance methods. The authors propose the use of fingerprint recognition technology as a secure and reliable means of identifying individuals, as fingerprints are unique to each user and difficult to replicate or forge. The system is intended for use in educational institutions and corporate

workplaces, where managing large numbers of employees or students manually can be both time-consuming and prone to errors. Upon registration, users' fingerprint data is securely stored in a local or cloud-based database. During check- in, the system scans the fingerprint and matches it with the stored templates to verify identity before logging attendance.The study concludes that integrating GPS and NFC technologies in attendance systems offers a practical, efficient, and secure solution for educational institutions aiming to modernize their administrative processes.

2.8 Development and Evaluation of an Attendance Tracking System Using Smartphones with GPS and NFC, Victor R.L. Shen, 2025

In the proposed system, students use their personal Android smartphones equipped with GPS and NFC functionalities. Upon entering the classroom, students tap their devices on an NFC tag, which records their attendance. Simultaneously, the application verifies the student's location using GPS to ensure they are within the authorized area. This two-step verification process enhances the reliability of attendance records. The system's design emphasizes user convenience, eliminating the need for additional hardware or infrastructure, as it utilizes devices that students already possess.The authors conducted experimental evaluations to assess the system's effectiveness. Results indicated a notable reduction in the time required for attendance tracking and an improvement in accuracy compared to traditional methods. Moreover, by utilizing students' own devices, the system minimizes costs associated with hardware procurement and maintenance.

2.9 Biometric Authentication Based Attendance System using GPS for Smart Phone , Praveenkumar Hiremath and Naveen Mirajkar, 2025

Biometric Authentication Based Attendance System Using GPS for Smartphone by Praveenkumar Hiremath and NaveenMirajkar presents a secure and intelligent approach to managing attendance by combining biometric authentication with real-time GPS location tracking through smartphones. The system is designed to address the persistent issues in traditional attendance methods such as proxy marking, manual errors, and inefficiency. In the system architecture described, the user registers through a mobile application by submitting personal details along with biometric data (fingerprint). During attendance, the application verifies the fingerprint using the phone's biometric scanner and simultaneously acquires the GPS coordinates of the user's location. These coordinates are then validated against a pre-configured geofence, such as a school, college, or office location. Only when both conditions are satisfied does the system allow attendance to be marked.

2.10 Wireless Fingerprint Attendance Management System, Varsha Karandikar et al(2025)

Wireless Fingerprint Attendance Management System, the authors present a sophisticated attendance system that integrates fingerprint biometrics, QR code scanning, and GPS location tracking. Designed to meet the demands of remote or hybrid working environments, the system allows users to mark their attendance wirelessly through mobile or dedicated devices. The inclusion of QR codes enables quick identity verification, while fingerprint authentication ensures that only authorized individuals can mark their presence. GPS tracking is used to validate the unauthorized locations. This three-layered authentication model significantly enhances the reliability and security of attendance records. The system stores all data in a centralized cloud database, providing real-time access for administrators to view, analyze, and generate reports. The paper highlights the

importance of mobility and wireless communication in the post- pandemic era, where flexibility and remote work have become standard. By combining multiple verification technologies and wireless connectivity, the proposed system offers a versatile, tamper-resistant, and scalable solution suitable for educational institutions, corporate offices, and remote team. Designed to meet the demands of remote or hybrid working environments, the system allows users to mark their attendance wirelessly through mobile or dedicated devices. This three-layered authentication model significantly enhances the reliability and security of attendance records and helps in wireless communication.

CHAPTER 3

SYSTEM ANALYSIS

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3.1 EXISTING SYSTEM

In many educational institutions and organizations, various attendance systems are currently in use, each with its own limitations. The most traditional method is the manual attendance system, where attendance is recorded by hand in registers or on paper. While simple, this method is time- consuming, prone to human error, and vulnerable to manipulation such as proxy marking. To overcome some of these issues, card-based or RFID (Radio Frequency Identification) systems have been introduced. These systems require individuals to scan an ID card on a reader to register their presence. Although faster and more automated than manual systems, they still pose security concerns as cards can be lost, stolen, or shared with others to mark fake attendance. Another common solution is the biometric attendance system using fingerprint or facial recognition.

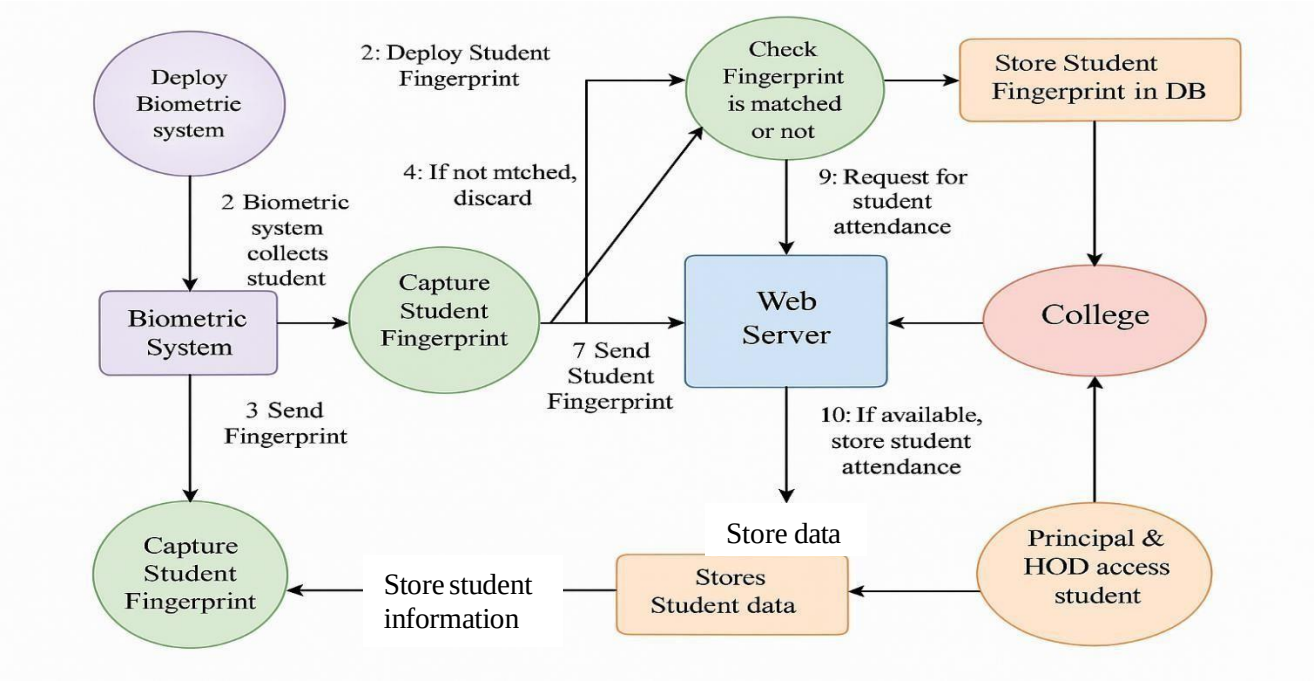

Figure 3.1 Existing System of Biometric Attendance

3.1.1 DISADVANTAGES

- Systems like manual entry or RFID cards do not confirm the actual identity of the person marking attendance, making them vulnerable to proxy or fraudulent entries.
- App-based or web-based systems without GPS do not verify if the user is physically present at the required location, allowing attendance from unauthorized places.
- Biometric systems installed at fixed locations are not suitable for remote work or field activities, limiting their use in dynamic environments.Hardware-based systems like RFID or biometric scanners require regular maintenance and come with installation and operational costs that can be burdensome for some organizations.

3.2 PROPOSED SYSTEM

The proposed system aims to provide a secure, efficient, and location-aware attendance management solution by integrating biometric authentication with GPS technology through a smartphone application. In this system, each user is required to verify their identity using the fingerprint sensor available on their smartphone. Once the biometric verification is successful, the application automatically captures the real-time GPS coordinates of the user's location. These coordinates are then matched against pre-defined geofenced areas, such as the campus or office premises. Attendance is recorded only when both conditions-biometric verification and correct location-are satisfied, ensuring that the user is physically present at the authorized site. This dual-layer authentication eliminates the possibility of proxy attendance and adds an extra layer of security. The attendance data is stored in a cloud database, which enables real-time access for administrators and facilitates the generation of reports and analytics.

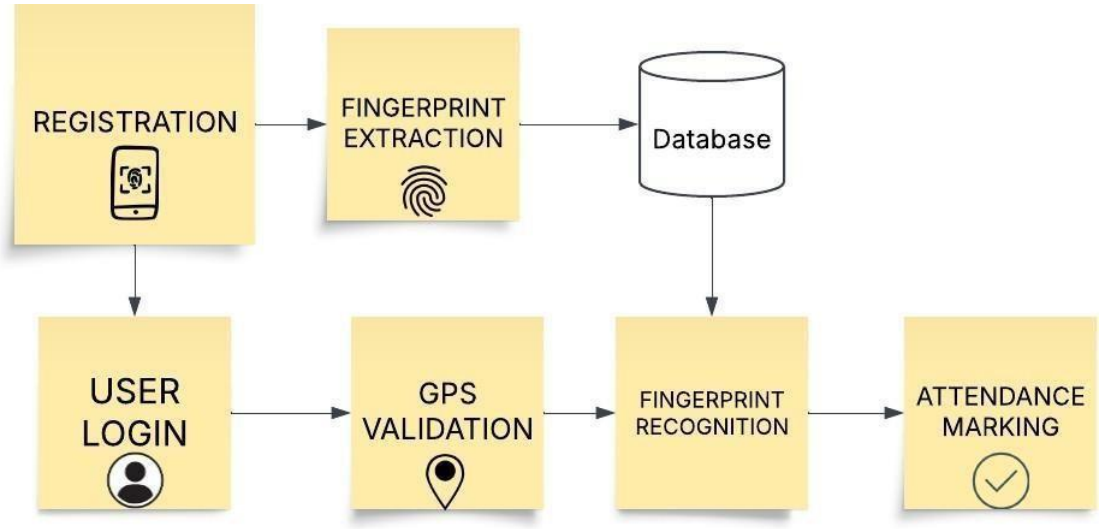

Figure3.2: Proposed System of Biometric Attendance

3.2.1 ADVANTAGES

- The combination of biometric authentication and GPS location verification ensures that only the authorized person at the correct location can mark attendance, eliminating proxy entries.
- The system captures live GPS coordinates, allowing organizations to verify the exact location of the user at the time of attendance, improving reliability and transparency.
- Since the system runs on smartphones, it supports attendance tracking from various locations-ideal for remote workers, field staff, and institutions without fixed biometric infrastructure.

3.3 SYSTEM CONFIGURATION

This biometric attendance app requires Android Studio (Arctic Fox or newer) with Java 8 or 11 support.The minimum system specs include Windows 10/Ubuntu 20.04+, 8 GB RAM, and Intel i5 or equivalent.Android SDK with API level 28 or higher is needed for BiometricPrompt API support.Google Play Services and location access must be enabled for GPS-based attendance.The app uses SQLite for local data and supports fingerprint-based login and attendance.Required permissions include biometric, location, internet, SMS, and call.Real device testing is recommended for fingerprint and location features.Gradle and dependencies like biometric and play-services-location must be properly configured.

3.3.1 Hardware Requirements

Component		Requirement
OS	-	Windows 10 / 11 (64-bit), macOS (Catalina or later)
RAM	-	8 GB
Processor	-	Intel i5/i7 or AMD Ryzen 5+ (with virtualization support)
Storage	-	At least 10 GB free (SSD recommended)
Graphics	-	Integrated or discrete GPU (OpenGL 3.0 or higher support)

3.3.2 Software Requirements

Software		Version
Android Studio	-	Arctic Fox (2020.3.1) or newer
Java SDK	-	Java 8 or 11
Gradle	-	7.0+
Android Gradle Plugin	-	7.0.0+
Emulator or Device	-	Android 9 (API 28) or higher

3.1 ARCHITECTURE DIAGRAM

A biometric attendance system using GPS and smartphones involves several key components: a smartphone application, a server, and a database. The smartphone app captures biometric data (fingerprint, face) and GPS location, which are then transmitted to the server for verification and storage. The server validates the data, stores it in the database, and can provide reports and analytics.

- **Biometric Data Capture:**

The user's smartphone application captures biometric data (e.g., fingerprint, facial recognition).

- **GPS Location Capture:**

The application also captures the user's current GPS location.

- **Biometric Verification:**

The server verifies the biometric data against a stored database of authorized users.

- **Location Validation:**

The server may also validate the GPS location to ensure it falls within an authorized area.

- **Data Storage:**

If the biometric and GPS data are valid, the attendance information (time, location, user ID) is stored in the database.

- **Attendance Records:**

Stores all attendance records, including timestamps, locations, and user IDs.

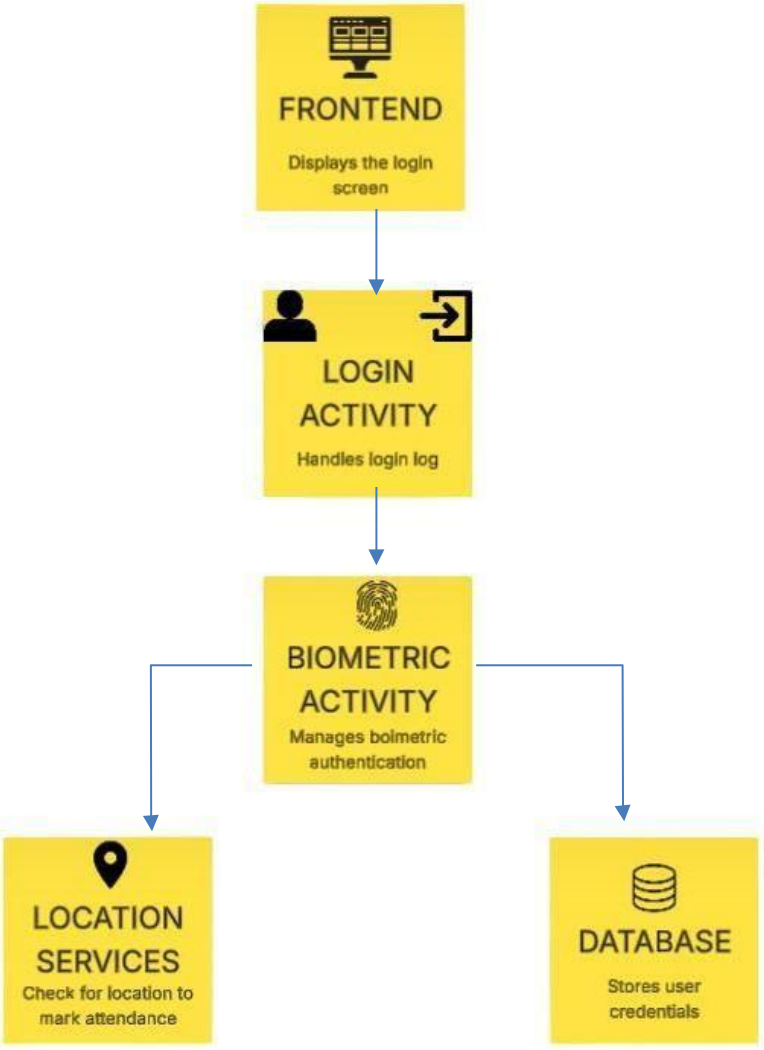

Figure3.3: Architecture Diagram of Biometric Attendance

CHAPTER 4

MODULES

4.1 MODULE DESCRIPTION

- User Registration
- User Login
- GPS Validation
- Fingerprint Authentication and Attendance Marking

4.1.1 User Registration

The Register module is the initial and foundational component of the Biometric Attendance Using Smartphone and GPS Coordinates system. This module is responsible for securely capturing and storing user credentials and biometric data, which are later used for identity verification during attendance marking. The purpose of this module is to ensure that only authorized users are allowed to access the system and record their attendance.

During the registration process, the user is required to enter their email address, create a secure password, and provide their fingerprint through the smartphone’s built-in biometric sensor. The email serves as the unique user identifier, while the password acts as an additional security layer. The fingerprint data is captured using the mobile device’s biometric API and securely stored in encrypted form within the system’s database or linked through device-level authentication, depending on the platform's security protocol.

To enhance security and prevent duplication, the system checks whether the provided email is already associated with an existing account. If not, the user’s credentials are securely stored in the cloud database.

This module plays a critical role in the overall system by enabling user authentication and access control. By integrating fingerprint verification at the time of registration, it sets the foundation for a tamper-proof attendance mechanism. Only users who have successfully registered can proceed to use the application's core features such as marking attendance or viewing attendance history.

4.1.2 User Loigin

The Login module is the second essential component of the Biometric Attendance Using Smartphone and GPS Coordinates system. It is responsible for authenticating registered users and granting them access to the system's features, including attendance marking and viewing attendance history. This module ensures that only authorized users who have successfully completed the registration process can log in and interact with the application.

In this module, users are required to enter their registered email address and password. To strengthen security, the system also prompts the user to authenticate via the fingerprint scanner of their smartphone. The use of biometric authentication in the login process adds a second layer of verification, ensuring that even if login credentials are compromised, unauthorized access is still prevented.

Once the user inputs their credentials, the system checks the provided information against the data stored in the cloud database. If the email and password match, and the fingerprint verification is successful, access is granted. Otherwise, the system displays an appropriate error message and prevents login.

4.1.3 GPS Validation

The GPS Validation module is a critical component of the Biometric Attendance Using Smartphone and GPS Coordinates system. It is responsible for verifying the physical presence of the user at a predefined, authorized location before allowing attendance to be marked. This ensures that attendance is recorded only when the user is genuinely present within the designated premises, adding a crucial layer of location-based authentication to the system.

After a successful login and biometric verification, the application uses the smartphone's built-in GPS functionality to obtain the real-time geographic coordinates (latitude and longitude) of the user. These coordinates are then compared with pre-stored location values representing the authorized attendance zones-such as a college campus, office building, or designated fieldwork area. A geofencing technique is used to define a virtual boundary around these approved zones. If the user's current GPS location falls within this boundary, they are allowed to proceed with marking attendance. Otherwise, the system denies access and notifies the user that they are outside the permitted area.

4.1.4 Fingerprint Authentication and Attendance Marking

The Fingerprint Authentication and Attendance Marking module is the final and most crucial component of the Biometric Attendance Using Smartphone and GPS Coordinates system. This module combines secure biometric identity verification with real-time attendance recording, ensuring that only the registered user, physically present at an authorized location, can successfully mark their attendance. When a user initiates the attendance process, the system first prompts for fingerprint authentication using the smartphone's built-in biometric sensor.

This fingerprint is verified against the fingerprint data securely linked to the user's profile during the registration phase. The authentication process is fast, accurate, and protected by the device's secure hardware, ensuring that the identity verification is both reliable and tamper-proof.

Once the fingerprint is successfully authenticated, the system proceeds to mark the attendance. It captures the current date, time, and GPS location, and checks whether the user is within the predefined geofenced area. This ensures that the user is physically present at the location where attendance is permitted, such as a campus or workplace. If the location is valid and authentication is successful, the attendance is marked and stored in the cloud-based database. If either the fingerprint fails or the user is outside the permitted area, the attendance marking is denied, and the user is notified.

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 FRONTEND

Android overview:

Android (stylized as android) is a mobile operating system developed by Google, based on the Linux kernel and designed primarily for touch screen mobile devices such as smart phones and tablets. Android's user interface is mainly based on direct manipulation, using touch gestures that loosely correspond to real-world actions, such as swiping, tapping and pinching, to manipulate on- screen objects, along with a virtual keyboard for text input. In addition to touch screen devices, Google has further developed Android TV for televisions, Android Auto for cars, and Android Wear for wrist watches, each with a specialized user interface. Variants of Android are also used on notebooks, game consoles, digital cameras, and other electronics. Android is a powerful Operating System supporting many applications in Smart Phones. These applications make life more comfortable and advanced for the users. Hardware's that support Android are mainly based on ARM architecture platform. Android comes with an Android market which is an online software store. It was developed by Google. It allows Android users to select, and download applications developed by third party developers and use them. There are around 2.0 lack+ games, application and widgets available on the market for users. Android applications are written in java programming language. Android is available as open source for developers to develop applications which can be further used for selling in android market. There are around 200000 applications developed for android with over 3billion+ downloads. Android relies on Linux version 2.6 for core system services such as security, memory management, process management, network stack.

5.2 BACKEND

Java Platform:

A platform is the hardware or software environment in which a program runs. The Java platform differs from most other platforms in that it's a software- only platform that runs on top of other, hardware-based platforms. Most other platforms are described as a combination of hardware and operating system.The Java platform has two components: The Java Virtual Machine (Java VM) and The Java Application Programming Interface (Java API) Java VM is the base for the Java platform and is ported onto various hardware-based platforms. The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries (packages) of related components. The following figure of Java Structure depicts a Java program, such as an application or applet, that's running on the Java platform. As the figure shows, the Java API and Virtual Machine insulates the Java program from hardware dependencies.

5.3 TOOL

Android SDK–API23

Android is an operating system based on Linux with a Java programming interface. The Android Software Development Kit(Android SDK) provides all necessary tools to develop Android applications. This includes a compiler, debugger and a device emulator, as well as its own virtual machine to run Android programs. Android is primarily developed by Google. Android allows background processing, provides a rich user interface library, supports2-D and 3-D graphics using the Open GL libraries, access to the file system and provides an embedded SQLite database. Android application consists of different components and can reuse components of other applications.

Android Development Tools

Android is a widely anticipated open-source operating system for mobile devices that provides a base operating system, an application middle ware layer, a Java software development kit (SDK), and a collection of system applications. Android mobile application development is based on Java language codes, as it allows developers to write codes in the Java language as illustrated in the below architecture figure of Android structure.

Android Studio13.2

Android Studio is the official Integrated Development Environment(IDE) for Android app development, based on IntelliJ IDEA. On top of IntelliJ's powerful code editor and developer tools, Android Studio offers even more features that enhance your productivity when building Android apps, such as: A flexible Gradle- based build system- A fast and feature-rich emulator- A unified environment where you can develop for all Android devices- Instant Run to push changes to your running app without building a new APK- Code templates and GitHub integration to help you build common app features and import sample code- Extensive testing tools and frameworks- Lint tools to catch performance, usability, version compatibility, and other problems-C++and NDK support-Built-in support for Google Cloud Platform, making it easy to integrate Google Cloud Messaging and App Engine.

CHAPTER 6

TEST RESULT AND ANALYSIS

6.1 TESTING

The testing phase of the Biometric Attendance Using Smartphone and GPS Coordinates project is critical to ensure the reliability, security, and functionality of the system before deployment. Comprehensive testing is conducted across all modules, including registration, login, attendance marking, and viewing attendance records. Functional testing is performed to verify that each feature works as expected, such as successful fingerprint verification, accurate GPS location capture, and proper database storage. Usability testing is carried out to evaluate the user interface and ensure that users can navigate the app easily without confusion. Security testing focuses on validating the robustness of user authentication, ensuring that data is encrypted and protected against unauthorized access or tampering. GPS testing is also crucial to confirm that the location tracking is accurate and that attendance is only recorded within the defined geofenced area. Additionally, integration testing is done to verify seamless communication between the front-end interface, biometric sensors, GPS module, and cloud database. Edge cases, such as incorrect fingerprints, GPS unavailability, or weak network conditions, are tested to assess how the system handles failures or exceptions. The outcome of the testing phase is a stable, secure, and user-friendly attendance management system that performs consistently across various devices and environments.

6.2 TEST OBJECTIVES

The primary objective of testing the Biometric Attendance Using Smartphone and GPS Coordinates project is to ensure the system performs reliably, securely, and accurately under various conditions. One of the key goals is to verify the functional correctness of each module, including registration, login, attendance marking, and viewing records, to confirm that they operate as intended. Special focus is given to validating the biometric authentication process, ensuring that only registered fingerprints are accepted and unauthorized access is effectively blocked. Another critical objective is to test the GPS-based location tracking, making sure the system captures accurate coordinates and permits attendance marking only within the predefined geofenced area. In addition, testing aims to evaluate the system's security by ensuring that sensitive user data, such as biometric information and location details, is stored and transmitted securely. The testing process also focuses on user experience by assessing the simplicity, responsiveness, and clarity of the interface across different devices. Integration testing is conducted to confirm smooth communication between biometric sensors, GPS modules, cloud storage, and the app interface. Lastly, the system is tested under various edge cases-such as weak network signals, invalid input, and denied permissions-to ensure that it handles exceptions gracefully without crashing or compromising data integrity. These objectives collectively aim to deliver a robust, user-friendly, and tamper-proof attendance system.

CHAPTER 7

RESULT AND DISCUSSION

7.1 RESULT

The testing and implementation of the Biometric Attendance Using Smartphone and GPS Coordinates system yielded highly satisfactory results, validating the reliability and effectiveness of the proposed solution. The registration module successfully recorded user credentials and biometric data, with no instances of duplication or data loss. During login testing, the system consistently authenticated users through both password and fingerprint verification, and rejected unauthorized access attempts accurately. The core functionality of the system-the mark attendance module-performed exceptionally well by correctly verifying user identity through fingerprint scans and capturing real-time GPS coordinates with a high degree of precision. Attendance was allowed only when the user was within the authorized location range, effectively preventing proxy attendance or manipulation. Additionally, the system demonstrated stable performance under various conditions, including low network environments and GPS delays, with minimal impact on functionality. The view attendance module accurately retrieved and displayed historical records, including date, time, and location, enabling users and administrators to track attendance seamlessly. Data was stored securely in the cloud and remained consistent across multiple sessions. The user interface was found to be intuitive and user-friendly during usability testing. Overall, the system fulfilled all the testing objectives and proved to be a reliable, secure, and efficient solution for modern attendance management using biometric and GPS technologies.

7.2 CONCLUSION

In conclusion, the Biometric Attendance Using Smartphone and GPS Coordinates project provides a modern, secure, and efficient solution for managing attendance by combining the strengths of biometric authentication and real-time GPS location tracking. The system effectively addresses the limitations of traditional attendance methods, such as proxy marking, lack of identity verification, and fixed infrastructure dependency. By utilizing fingerprint verification and capturing GPS coordinates through a user-friendly mobile application, the system ensures that attendance is marked only when the user is physically present at an authorized location. The integration with a cloud-based database further enhances data security, accessibility, and scalability. The successful implementation and testing of all modules demonstrate the reliability and practicality of this approach in both academic and professional settings. Overall, the system stands as a robust and innovative solution, paving the way for smarter, technology-driven attendance management.

7.3 FUTURE ENHANCEMENT

In the future, the Biometric Attendance Using Smartphone and GPS Coordinates system can be enhanced with several advanced features to increase its flexibility and usability. One key improvement is the implementation of multiple user logins on the same device, which would be particularly beneficial in shared environments such as classrooms, offices, or laboratories where multiple individuals use a single smartphone or tablet. This feature would allow each user to securely log in with their own credentials and fingerprint without interfering with other users' data, making the system more versatile and practical in real- world scenarios. Additionally, future updates could include facial recognition as an alternative biometric method, offline attendance marking with delayed synchronization, customizable geofencing boundaries for different institutions, and integration with third-party platforms like HR or academic management systems.

APPENDIX- A

SOURCE CODE

ProfileActivity.java

```
package com.example.biometricattendance;
import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationManager;
import android.os.Bundle;
import android.provider.Settings;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;
import java.util.concurrent.Executor;
public class ProfileActivity extends AppCompatActivity {
    TextView emailText, attendanceText;
    Button markAttendanceBtn, viewRecordsBtn, logoutBtn;
    DatabaseHelper dbHelper;
    String userEmail;
    // Predefined allowed location (e.g., office/school)
    private static final double ALLOWED_LAT = 12.9716;
```

```
    private static final double ALLOWED_LNG = 77.5946;
    private static final float LOCATION_TOLERANCE_METERS = 100f; // 100
meters

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_profile);
        emailText = findViewById(R.id.profile_email);
        attendanceText = findViewById(R.id.attendance_count);
        markAttendanceBtn = findViewById(R.id.mark_attendance_btn);
        viewRecordsBtn = findViewById(R.id.view_records_btn);
        logoutBtn = findViewById(R.id.logout_btn);
        dbHelper = new DatabaseHelper(this);
        userEmail = getIntent().getStringExtra("email");
        emailText.setText("Welcome: " + userEmail);
        updateAttendanceCount();
        markAttendanceBtn.setOnClickListener(v ->
checkLocationAndMarkAttendance());
        viewRecordsBtn.setOnClickListener(v -> {
            Intent intent = new Intent(ProfileActivity.this,
AttendanceRecordsActivity.class);
            intent.putExtra("email", userEmail);
            startActivity(intent);
        });
        logoutBtn.setOnClickListener(v -> {
            Intent intent = new Intent(ProfileActivity.this, LoginActivity.class);
            startActivity(intent);
            finish();
        });
    }
```

```
    private void updateAttendanceCount() {
        int count = dbHelper.getAttendanceCount(userEmail);
        attendanceText.setText("Total Present Days: " + count);
    }

    private void checkLocationAndMarkAttendance() {
        LocationManager locationManager = (LocationManager)
getSystemService(LOCATION_SERVICE);
        boolean isGPSEnabled =
locationManager.isProviderEnabled(LocationManager.GPS_PROVIDER);
        boolean isNetworkEnabled =
locationManager.isProviderEnabled(LocationManager.NETWORK_PROVIDER);
        if (!isGPSEnabled && !isNetworkEnabled) {
            Toast.makeText(this, "Location services are disabled. Please turn on GPS.",
Toast.LENGTH_LONG).show();
           startActivity(new
Intent(Settings.ACTION_LOCATION_SOURCE_SETTINGS));
            return;
        }
        Location location = null;
        if (ActivityCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION) ==
PackageManager.PERMISSION_GRANTED) {
            if (isGPSEnabled)
                location =
locationManager.getLastKnownLocation(LocationManager.GPS_PROVIDER);
            if (location == null && isNetworkEnabled)
                location =
locationManager.getLastKnownLocation(LocationManager.NETWORK_PROVID
ER);
        } else {
```

```
            Toast.makeText(this, "Location permission not granted",
Toast.LENGTH_SHORT).show();
            return;
        }
        if (location != null) {
            double currentLat = location.getLatitude();
            double currentLng = location.getLongitude();
            float[] distance = new float[1];
            Location.distanceBetween(currentLat, currentLng, ALLOWED_LAT,
ALLOWED_LNG, distance);
            if (distance[0] < LOCATION_TOLERANCE_METERS) {
                authenticateAndMark("Present");
            } else {
                markAttendance("absent");
                Toast.makeText(this, "marked as absent",
Toast.LENGTH_SHORT).show();
                updateAttendanceCount();
            }
        } else {
            Toast.makeText(this, "unable to get the location",
Toast.LENGTH_SHORT).show();
        }
    }
    private void authenticateAndMark(String statusToMark) {
        Executor executor = ContextCompat.getMainExecutor(this);
        BiometricPrompt biometricPrompt = new
BiometricPrompt(ProfileActivity.this, executor,
                new BiometricPrompt.AuthenticationCallback() {
                    @Override
                    public void onAuthenticationSucceeded(@NonNull
BiometricPrompt.AuthenticationResult result) {
```

```
                super.onAuthenticationSucceeded(result);
                markAttendance(statusToMark);
                Toast.makeText(ProfileActivity.this, "Attendance marked: " +
statusToMark, Toast.LENGTH_SHORT).show();
                updateAttendanceCount();
            }
            @Override
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(ProfileActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
    BiometricPrompt.PromptInfo promptInfo = new
BiometricPrompt.PromptInfo.Builder()
        .setTitle("Confirm Attendance")
        .setSubtitle("Scan fingerprint to mark " + statusToMark)
        .setNegativeButtonText("Cancel")
        .build();
    biometricPrompt.authenticate(promptInfo);
  }
  private void markAttendance(String status) {
    String date = new SimpleDateFormat("yyyy-MM-dd",
Locale.getDefault()).format(new Date());
    String time = new SimpleDateFormat("HH:mm:ss",
Locale.getDefault()).format(new Date());
    dbHelper.markAttendance(userEmail, date, time, status);
  }
}
```

activity_profile.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/profile_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#FFFFFF">
    <TextView
        android:id="@+id/profile_email
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome: user@example.com"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp" />
    <TextView
        android:id="@+id/attendance_count"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Total Present Days: 0"
        android:textSize="16sp"
        android:layout_marginBottom="32dp" />
    <Button
        android:id="@+id/mark_attendance_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Mark Attendance"
```

```
        android:layout_marginBottom="16dp"
        android:backgroundTint="#4CAF50"
        android:textColor="#FFFFFF" />
    <Button
        android:id="@+id/view_records_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Attendance Records"
        android:layout_marginBottom="16dp"
        android:backgroundTint="#2196F3"
        android:textColor="#FFFFFF" />

    <Button
        android:id="@+id/logout_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Logout"
        android:backgroundTint="#F44336"
        android:textColor="#FFFFFF" />
</LinearLayout>
```

RegisterActivity.java

```
// RegisterActivity.java
package com.example.biometricattendance;
import android.annotation.SuppressLint;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.content.ContextCompat;
import java.util.concurrent.Executor;
public class RegisterActivity extends AppCompatActivity {
    EditText etEmail, etPassword;
    Button btnRegister;
    DatabaseHelper databaseHelper;
    BiometricPrompt biometricPrompt;
    BiometricPrompt.PromptInfo promptInfo;
    String email = "";
    String password = "";
    String generatedFingerprint = "";
    @SuppressLint("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
       etEmail = findViewById(R.id.et_register_email);
        etPassword = findViewById(R.id.et_register_password);
        btnRegister = findViewById(R.id.btn_register);
        databaseHelper = new DatabaseHelper(this);
        setupBiometricPrompt();
        btnRegister.setOnClickListener(v -> {
            email = etEmail.getText().toString().trim();
            password = etPassword.getText().toString().trim();
            if (email.isEmpty() || password.isEmpty()) {
                Toast.makeText(this, "Fill all fields", Toast.LENGTH_SHORT).show();
                return;
            }
            if (databaseHelper.isEmailRegistered(email)) {
```

```
                Toast.makeText(this, "Email already registered",
Toast.LENGTH_SHORT).show();
                return;
            }
            biometricPrompt.authenticate(promptInfo);
        });
    }
    private void setupBiometricPrompt() {
        Executor executor = ContextCompat.getMainExecutor(this);
        biometricPrompt = new BiometricPrompt(this, executor, new
BiometricPrompt.AuthenticationCallback() {
            @Override
        public void
onAuthenticationSucceeded(BiometricPrompt.AuthenticationResult result) {
                super.onAuthenticationSucceeded(result);
                generatedFingerprint = email + "_fingerprint"; // Unique fingerprint
binding
                boolean success = databaseHelper.registerUser(email, password,
 generatedFingerprint);
                if (success) {
                    Toast.makeText(RegisterActivity.this, "Registration Successful",
Toast.LENGTH_SHORT).show();
                    startActivity(new Intent(RegisterActivity.this, LoginActivity.class));
                    finish();
                } else {
                    Toast.makeText(RegisterActivity.this, "Registration Failed",
Toast.LENGTH_SHORT).show();
                }
            }

            @Override
```

```
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(RegisterActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
        promptInfo = new BiometricPrompt.PromptInfo.Builder()
                .setTitle("Fingerprint Registration")
                .setSubtitle("Scan your fingerprint to register")
                .setNegativeButtonText("Cancel")
                .build();
    }
}
```

activity_register.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">
    <EditText
        android:id="@+id/et_register_email"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:padding="10dp" />
    <EditText
        android:id="@+id/et_register_password"
```

```
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:padding="10dp"
        android:layout_marginTop="16dp" />
    <Button
        android:id="@+id/btn_register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="24dp" />
</LinearLayout>
```

LoginActivity.java

```
package com.example.biometricattendance;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.biometric.BiometricPrompt;
import androidx.core.content.ContextCompat;
import java.util.concurrent.Executor;
public class LoginActivity extends AppCompatActivity {
    EditText emailInput, passwordInput;
    Button loginBtn;
    DatabaseHelper dbHelper;
    String enteredEmail;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);
        emailInput = findViewById(R.id.login_email);
        passwordInput = findViewById(R.id.login_password);
        loginBtn = findViewById(R.id.login_button);
        dbHelper = new DatabaseHelper(this);
        loginBtn.setOnClickListener(view -> {
            enteredEmail = emailInput.getText().toString();
            String password = passwordInput.getText().toString();
            if (dbHelper.checkUser(enteredEmail, password)) {
                authenticateFingerprint();
            } else {
                Toast.makeText(LoginActivity.this, "Invalid email or password",
Toast.LENGTH_SHORT).show();
            }
        });
    }
    private void authenticateFingerprint() {
        Executor executor = ContextCompat.getMainExecutor(this)
        BiometricPrompt biometricPrompt = new
BiometricPrompt(LoginActivity.this, executor,
                new BiometricPrompt.AuthenticationCallback() {
                    @Override
                    public void
onAuthenticationSucceeded(BiometricPrompt.AuthenticationResult result) {
                        super.onAuthenticationSucceeded(result);
                        Toast.makeText(LoginActivity.this, "Fingerprint Verified!",
Toast.LENGTH_SHORT).show();
```

```
                Intent intent = new Intent(LoginActivity.this, ProfileActivity.class);
                intent.putExtra("email", enteredEmail);
                startActivity(intent);
                finish();
            }
            @Override
            public void onAuthenticationFailed() {
                super.onAuthenticationFailed();
                Toast.makeText(LoginActivity.this, "Fingerprint not recognized",
Toast.LENGTH_SHORT).show();
            }
        });
    BiometricPrompt.PromptInfo promptInfo = new
BiometricPrompt.PromptInfo.Builder()
        .setTitle("Login Authentication")
        .setSubtitle("Scan your fingerprint to login")
        .setNegativeButtonText("Cancel")
        .build()
    biometricPrompt.authenticate(promptInfo);
  }
}
```

Activity_Login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="24dp">
  <TextView
```

```
        android:text="Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />
    <EditText
        android:id="@+id/login_email"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:layout_marginBottom="16dp"/>
    <EditText
        android:id="@+id/login_password"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:layout_marginBottom="24dp"/>
    <Button
        android:id="@+id/login_button"
        android:text="Login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
</LinearLayout>
```

AttendanceRecordsActivity.java

```
package com.example.biometricattendance;
import android.database.Cursor;
```

```
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
public class AttendanceRecordsActivity extends AppCompatActivity {
    ListView recordsListView;
    DatabaseHelper dbHelper;
    String currentEmail;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_attendance_records);
        recordsListView = findViewById(R.id.recordsListView);
        dbHelper = new DatabaseHelper(this);
        currentEmail = getIntent().getStringExtra("email");
        if (currentEmail == null) {
            Toast.makeText(this, "No user email provided.",
Toast.LENGTH_SHORT).show();
            return;
        }
        loadAttendanceRecords();
    }
    private void loadAttendanceRecords() {
        Cursor cursor = dbHelper.getAllAttendanceRecords(currentEmail);
        ArrayList<String> records = new ArrayList<>();
        if (cursor != null && cursor.moveToFirst()) {
            do {
                String date = cursor.getString(cursor.getColumnIndexOrThrow("date"));
                String time = cursor.getString(cursor.getColumnIndexOrThrow("time"));
```

```
            String status =
cursor.getString(cursor.getColumnIndexOrThrow("status"));
            records.add(date + " " + time + " - " + status);
        } while (cursor.moveToNext());
        cursor.close();
    } else {
        records.add("No attendance records found.");
    }
    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, records);
    recordsListView.setAdapter(adapter);
  }
}
```

activity_attendance_records.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">
<ListView
    android:id="@+id/recordsListView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:dividerHeight="1dp"/>
</LinearLayout>
```

DatabaseHelper.java

```
package com.example.biometricattendance;
import android.content.ContentValues;
```

```
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
public class DatabaseHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "AttendanceDB.db";
    private static final int DATABASE_VERSION = 1;
    // Users Table
    private static final String TABLE_USERS = "users";
    private static final String COLUMN_EMAIL = "email";
    private static final String COLUMN_PASSWORD = "password";
    private static final String COLUMN_FINGERPRINT = "fingerprint_data"; //
stub column for uniqueness
    private static final String TABLE_ATTENDANCE = "attendance";
    private static final String COLUMN_ID = "id";
    private static final String COLUMN_DATE = "date";
    private static final String COLUMN_TIME = "time";
    private static final String COLUMN_STATUS = "status";
    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }
    @Override
    public void onCreate(SQLiteDatabase db) {
        String createUsersTable = "CREATE TABLE " + TABLE_USERS + " (" +
                COLUMN_EMAIL + " TEXT PRIMARY KEY, " +
                COLUMN_PASSWORD + " TEXT, " +
                COLUMN_FINGERPRINT + " TEXT UNIQUE)";
        String createAttendanceTable = "CREATE TABLE " +
TABLE_ATTENDANCE + " (" +
                COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
+
```

```
            COLUMN_EMAIL + " TEXT, " +
            COLUMN_DATE + " TEXT, " +
            COLUMN_TIME + " TEXT, " +
            COLUMN_STATUS + " TEXT)";
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_USERS);
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ATTENDANCE);
        onCreate(db);
    }
    public boolean registerUser(String email, String password, String
fingerprintData) {
        if (isFingerprintRegistered(fingerprintData)) return false;
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(COLUMN_EMAIL, email);
        values.put(COLUMN_PASSWORD, password);
        values.put(COLUMN_FINGERPRINT, fingerprintData);
        long result = db.insert(TABLE_USERS, null, values);
        return result != -1;
    }
    public boolean checkUser(String email, String password) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_USERS, null, COLUMN_EMAIL + "=?
AND " + COLUMN_PASSWORD + "=?",
            new String[]{email, password}, null, null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) cursor.close();
        return exists;
    }
    public boolean isFingerprintRegistered(String fingerprintData) {
```

```
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_USERS, null, COLUMN_FINGERPRINT
+ "=?",
                new String[]{fingerprintData}, null, null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) cursor.close();
        return exists;
    }
    public boolean markAttendance(String email, String date, String time, String
status) {
        // Check if already marked for the date
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query(TABLE_ATTENDANCE, null,
                COLUMN_EMAIL + "=? AND " + COLUMN_DATE + "=?",
                new String[]{email, date}, null, null, null);
        if (cursor != null && cursor.moveToFirst()) {
            cursor.close();
            return false; // Already marked
        }
        ContentValues values = new ContentValues();
        values.put(COLUMN_EMAIL, email);
        values.put(COLUMN_DATE, date);
        values.put(COLUMN_TIME, time);
        values.put(COLUMN_STATUS, status);
        long result = db.insert(TABLE_ATTENDANCE, null, values);
        return result != -1;
    }
    public int getAttendanceCount(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery("SELECT COUNT(*) FROM " +
TABLE_ATTENDANCE +
```

```
            " WHERE " + COLUMN_EMAIL + "=? AND " + COLUMN_STATUS
+ "=?", new String[]{email, "Present"});
        int count = 0;
        if (cursor.moveToFirst()) {
            count = cursor.getInt(0);
        }
        cursor.close();
        return count;
    }
public Cursor getAttendanceRecords(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.query(TABLE_ATTENDANCE, null, COLUMN_EMAIL + "=?",
                new String[]{email}, null, null, COLUMN_DATE + " DESC");
    }
    public boolean isEmailRegistered(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.query("users", null, "email=?", new String[]{email}, null,
null, null);
        boolean exists = (cursor != null && cursor.moveToFirst());
        if (cursor != null) {
            cursor.close();
        }
        return exists;
    }
    public Cursor getAllAttendanceRecords(String email) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.query("attendance", null, "email=?", new String[]{email}, null, null,
"date DESC");
    }
}
```

APPENDIX- B

SCREENSHOTS

Sample Output

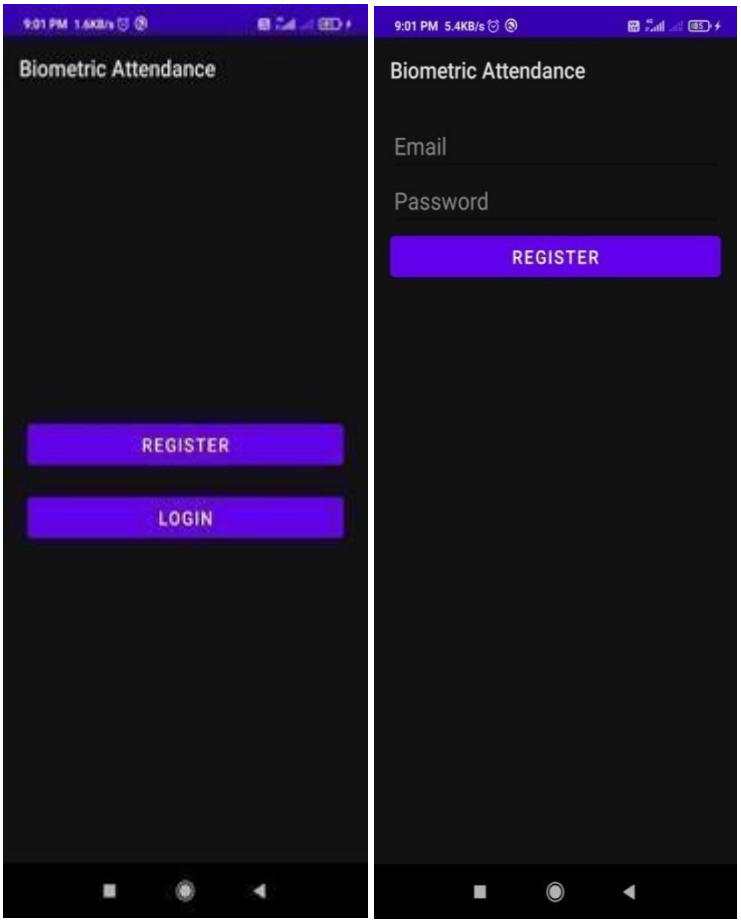

Figure B.1: User Registration

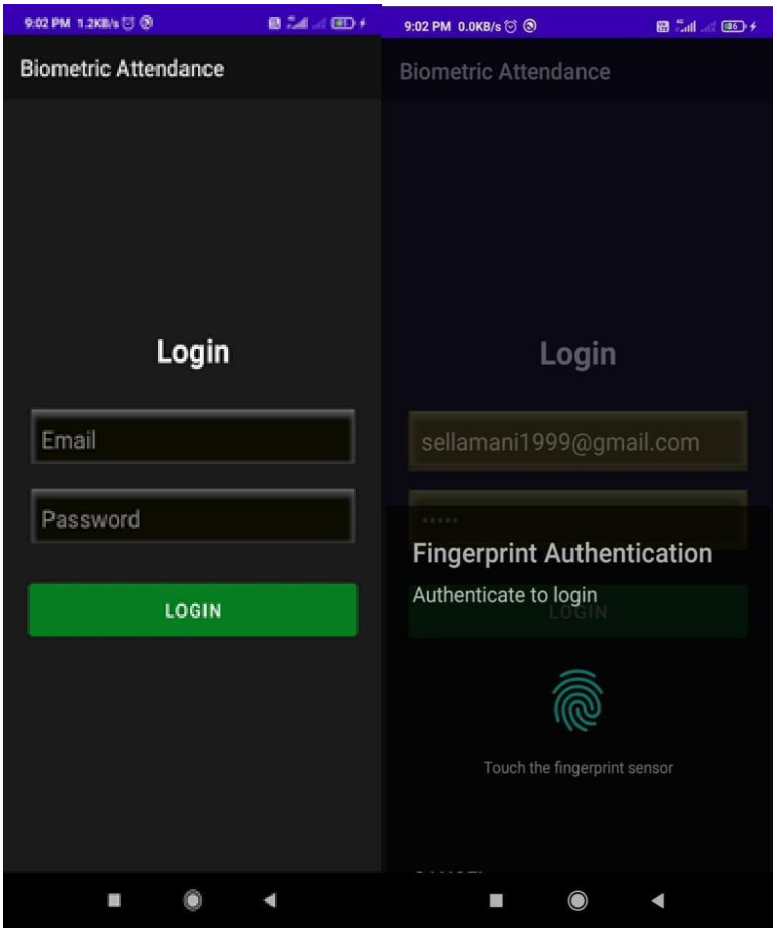

Figure B.2:User Login & Fingerprint Authentication

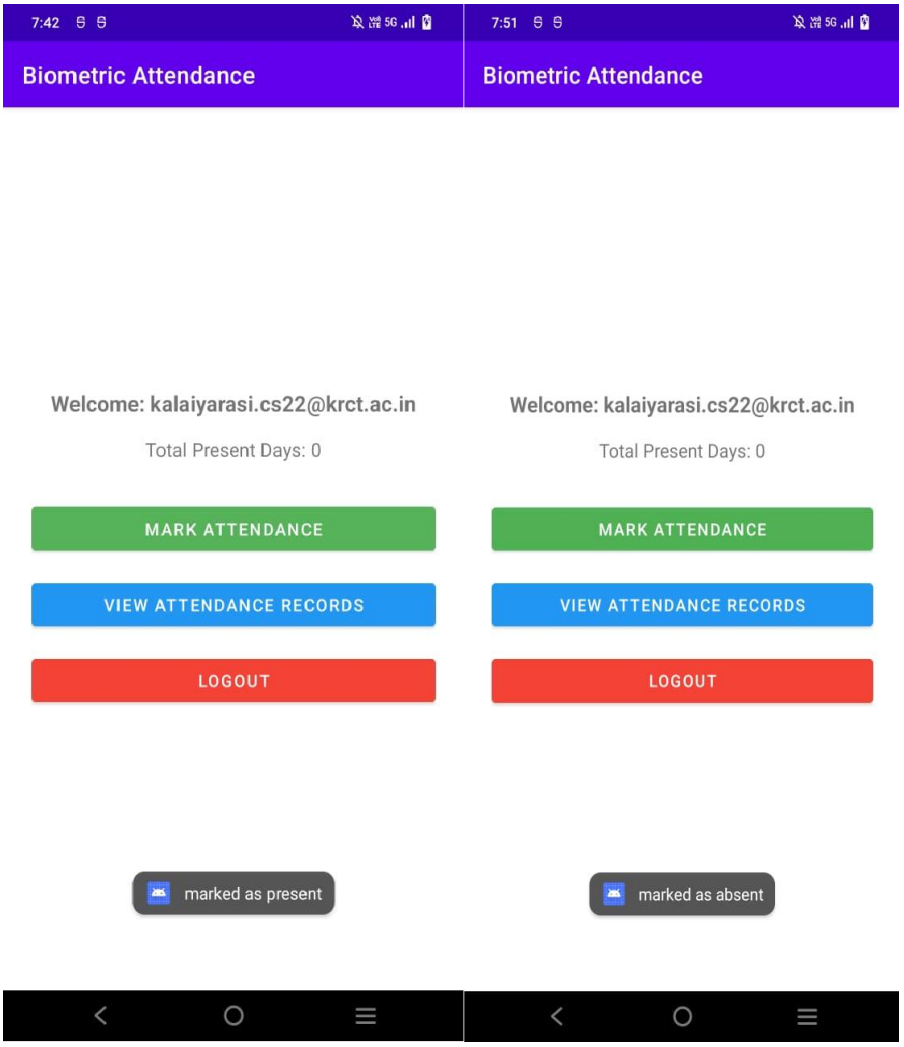

Figure B.3:Attendance marking

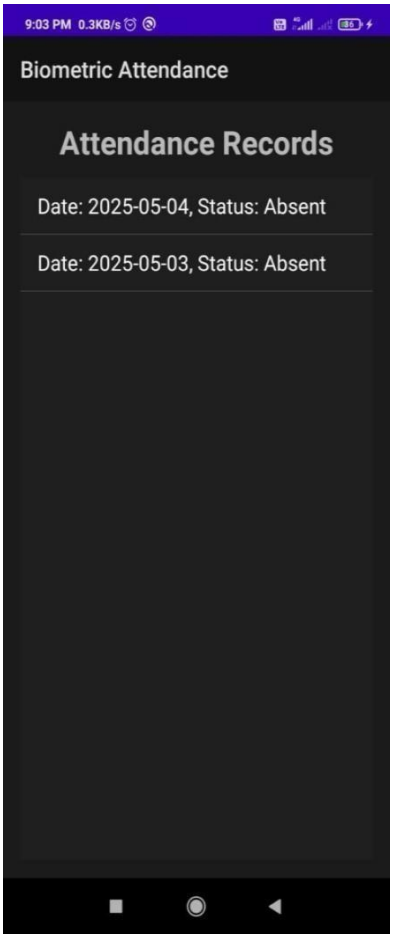

Figure B.4:Attendance Record

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